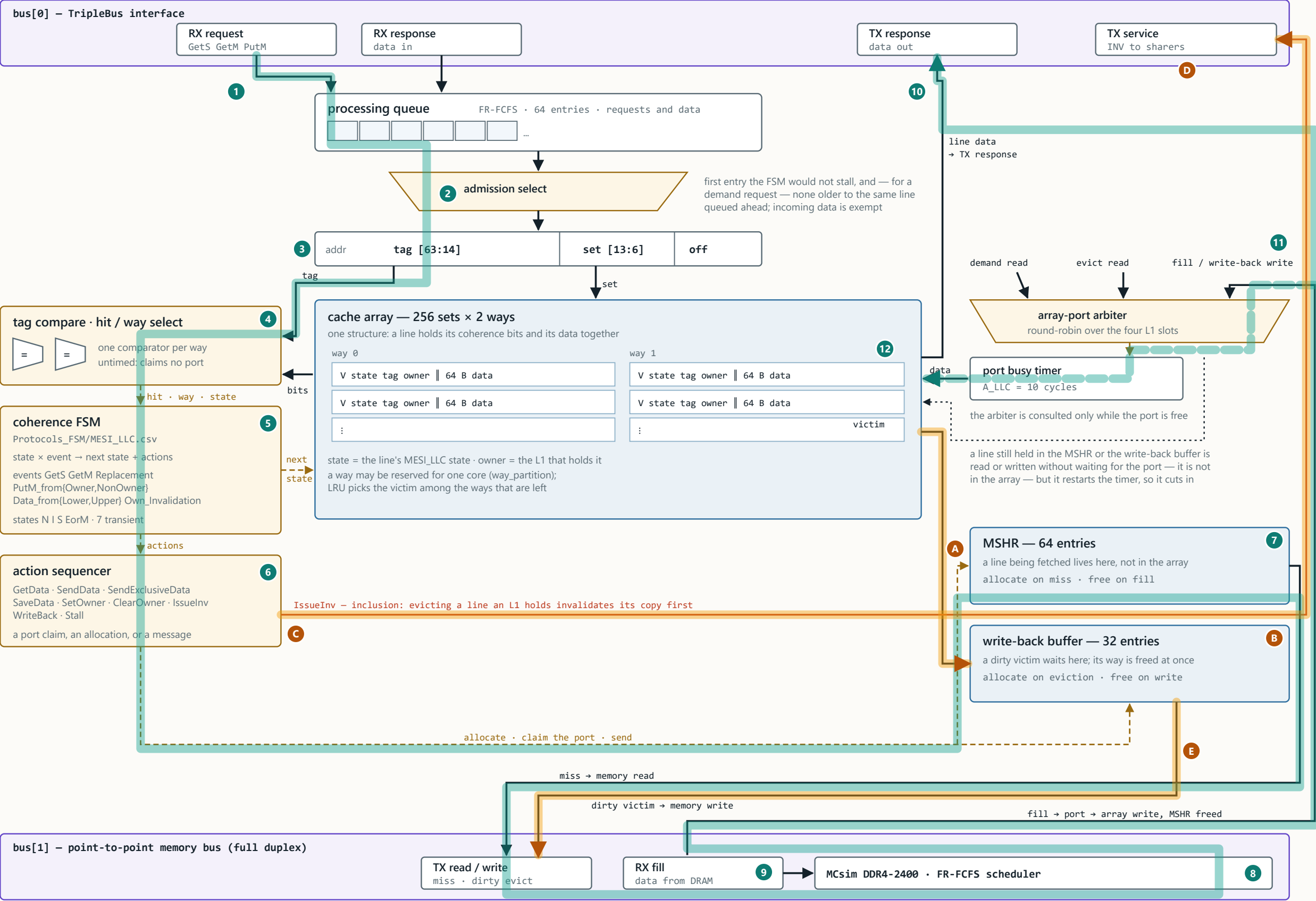


Octopus LLC datapath — the path of a read miss that evicts

CacheController_End2End + CacheDataHandler_COTS · 32 KB, 2-way, 64 B lines · MESI_LLC coherence FSM · one timed data port
Highlighted: a GetS missing in a FULL set — teal the requested line, amber the victim, chosen by LRU only when the fill's array write runs (dashed: off the critical path). Its INV goes out on TX service and it is always written back (no dirty bit). A GetM evicts identically; an owned victim returns its data first.



— data path (timed) register-resident access, bypasses the port - - - - - control (untimed) — back-invalidation (inclusion)

□ storage □ control logic □ interconnect interface the Logger and the raw trace observe: queue arrival · admission · port claim · data emitted

teal the requested line — 1 arrives · 2 admitted · 3 decoded · 4 tag miss; the set is full · 5 FSM: GetData, SetOwner · 6 sequencer: allocate, send · 7 MSHR · 8 DRAM read · 9 fill on RX · 10 response out, from the fill itself · off the critical path: 11 port claim → busy timer · 12 written into the victim's way, MSHR freed

victim — A to the write-back buffer as the fill lands (LRU among the requester's ways) · B Replacement raised · C IssueInv → INV on TX service · D sharers drop it; the LLC gets its own INV back · E WriteBack: read via the bypass, written to memory on bus[1]